

CSE-350 Quiz-03

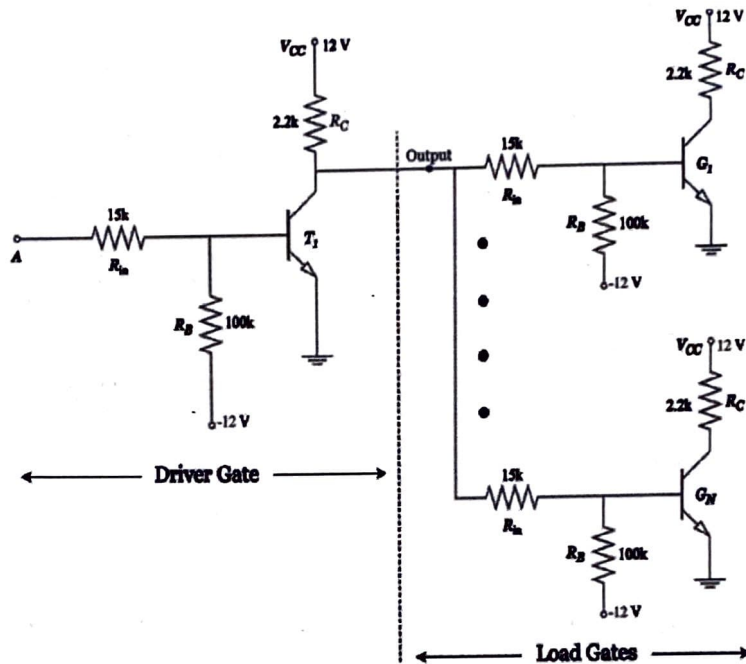
DL and RTL

Set-A

Total - 20 marks

Time - 25 minutes

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Problem-1		
10 marks		



a) For the given RTL circuit find maximum fanout. Given $V_{OH}=10V$, $V_{OL}=0.2V$, and $\beta_F=30$	5
b) Find the Noise Margin of the Driver Gate. Given $V_{OH}=10V$, $V_{OL}=0.2V$, and $\beta_F=30$	5

$$(a) I_S = \frac{12 - 10}{2.2} = 0.91 \text{ mA}$$

$$I_D = \frac{10 - 0.2}{15} = 0.613 \text{ mA}$$

$$\begin{aligned} \text{Fanout} &= \text{floor} \left(\frac{I_S}{I_D} \right) \\ &= \text{floor} \left(\frac{0.91}{0.613} \right) \\ &= \text{floor}(1.48) \\ &\approx 1 \end{aligned}$$

$$\begin{aligned} \therefore \text{Maximum fanout} &= \text{mm}(1, 50) \\ &= 1 \text{ (Ans).} \end{aligned}$$

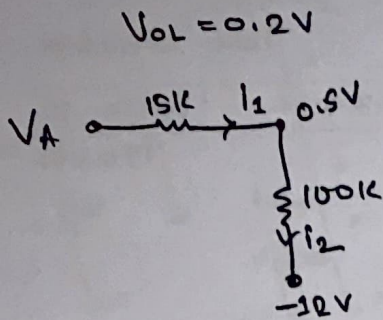
$$V_{OL} = 0.2V$$

$$I_S = \frac{12 - 0.2}{2.2} = 5.36 \text{ mA}$$

$$I_D = \frac{0.2 - (-12)}{15 + 100} = 0.106 \text{ mA}$$

$$\begin{aligned} \text{Fanout} &= \text{floor} \left(\frac{I_S}{I_D} \right) \\ &= \text{floor}(50.566) \\ &\approx 50. \end{aligned}$$

(b) Noise Margin = ? when $V_{OH} = 10V$, $V_{OL} = 0.2V$ and $\beta_F = 30$.



At cut-off so
 $V_{BE} = 0.5V$.

$$V_{BE} = V_B - V_E$$

$$0.5 = V_B - 0$$

$$\Rightarrow V_B = 0.5V$$

$$I_1 = I_2$$

$$\Rightarrow \frac{V_A - 0.5}{15} = \frac{0.5 - (-12)}{100}$$

$$\Rightarrow 100(V_A - 0.5) = 15(12.5)$$

$$\Rightarrow 100V_A - 50 = 187.5$$

$$\Rightarrow V_A = 2.375$$

$$\text{So, } V_{IL} = 2.375V$$

$$V_{NH} = V_{OH} - V_{IH}$$

$$= 10 - 5.402 = 4.598V$$

$$V_{NL} = V_{IL} - V_{OL}$$

$$= 2.375 - 0.2$$

$$= 2.175V$$

$$\therefore V_{NM} = \min(V_{NH}, V_{NL})$$

$$= \min(4.598, 2.175)$$

$$= 2.175V \text{ (Ans.)}$$

$$V_{OH} = 10V$$

In saturation,

$$\text{So, } V_{BE} = 0.8V,$$

$$V_{CE} = 0.2V.$$

$$I_1 = \frac{V_A - 0.8}{15}$$

$$I_2 = \frac{0.8 - (-12)}{100}$$

$$I_C = \frac{12 - 0.2}{2.2} = 5.364mA$$

$$\beta_F = \frac{I_C}{I_B}$$

$$\Rightarrow I_B = \frac{I_C}{\beta_F} = \frac{5.364}{30}$$

$$= 0.1788mA$$

$$\text{So, } I_1 = I_2 + I_B$$

$$\Rightarrow \frac{V_A - 0.8}{15} = \frac{0.8 + 12}{100} + 0.1788$$

$$\Rightarrow \frac{V_A - 0.8}{15} = \frac{0.8 + 12 + 17.88}{100}$$

$$\Rightarrow 100(V_A - 0.8) = 460.2$$

$$\Rightarrow 100V_A - 80 = 460.2$$

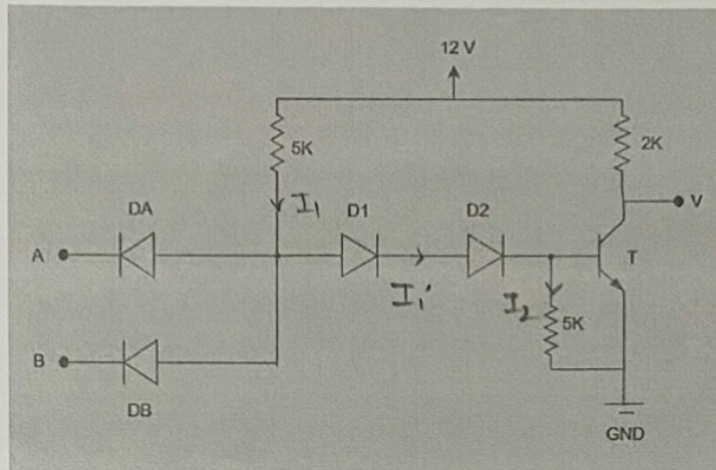
$$\Rightarrow V_A = \frac{460.2 + 80}{100}$$

$$= 5.402V$$

$$\therefore V_{IH} = 5.402V$$

Problem-2

10 marks



- (a) Calculate maximum power dissipation for the above circuit. ($V_{IL} = 0.2V$, $V_{IH} = 12V$, $V_Y = 0.5V$, $\beta_F = 30$)

a) max power dissipation when $V_A = V_B = 12V$

D_A, D_B both RB, $T \rightarrow \text{sat}$ $V_{BE} = 0.8V$

$$I_{DB} = I_{DA} = 0 \text{ mA}$$

$$V_{CE} = 0.2V$$

$$V_p = 0.7 + 0.7 + 0.8 = 2.2V$$

$$I_1 = \frac{12 - 2.2}{5} = 1.96 \text{ mA}$$

$$I_1 = I_1' = 1.96 \text{ mA}$$

$$I_2 = \frac{0.8 - 0}{5} = 0.16 \text{ mA}$$

$$I_e = \frac{12 - 0.2}{2} = 5.9 \text{ mA}$$

$$I_B = I_1' - I_2 = 1.96 - 0.16 = 1.8 \text{ mA}$$

$$P = (12 - 0)(I_2 + I_B + I_e)$$

$$= 12(0.16 + 1.8 + 5.9)$$

$$= 94.32 \text{ mWatt}$$